

1) Computer vision is a branch of Artificial Intelligence (AI). That enables computer to capture, process, analyze and interpret image & videos.

- Detect and Recognize objects.
- Tracks object of image & video.

b) Face Recognize System.

Autonomous self driving.

2) Key features included image acquisition, preprocessing, Feature Extraction, object detection, object Recognize, image segmentation, Pattern Recognize.

Object Recognize allows image by the objects accurately.

Low level image processing basic operation like noise removal, image contrast adjustment, blurring, image enhancement.



mid level :

- Focused on Segmentation and Feature Extraction.
- produce image features

High level :

- Focused on Object Recognizing & image understanding.
- produces meaningful decisions.

4) digital image is represented as matrix of pixels arranged in rows & columns.

A pixel is the smallest unit in digital image that stores color and intensity information.

Higher resolution means more pixels, images are clear & quality. Low resolution reduces image quality.

5) Light reflected from object is captured by image sensor converted into electrical signal.

into then transformed into digital image.

Capture light & convert into electrical signal.

into then transformed into digital image.

- Digital camera.

- Web cam.



- 2) Sampling : Sampling is a process of converting a continuous image into discrete pixels.
- b) Quantization assigns a finite number of intensity value to sampled pixels, enabling digital image representation.

7) The image formation model. Explains how light from an object passes through a camera lens and forms image on the sensor.

Perspective projection is the process of projecting 3D into 2D image plane while preserving depth.

Illumination is the provided light reflects from objects making them visible image sensor.

8) Image processing is the technique of enhancing, storing & modifying the images by using computer algorithm.

Computer enhanced the images output is an improved image.

Computer vision understand & interprets image content output is information.



92

Computer vision helps detect diseases such as Tumors and pneumonia by an analyzing machine images like x-rays, CT, MRIs.

It detect road signals, traffic signals, pedestrians & other vehicles, helping self driving cars navigate safely.

101

Digital image fundamentals provides the basic accurate image analysis, object detection & Recognition.

- a) Pixels stores brightness & color information which together form the complete image.
- b) Sampling determines the number of pixels while image resolution affects images clarity & detail. Higher sampling & Resolution improved accuracy of computer vision system.